



Triangles from Rectangles Teaching Bundle

Grade 6 • North Carolina • Mathematics

NC.6.G.1 • Use decomposition to find triangle area.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Use decomposition to find triangle area. This bundle practices a focused part of NC.6.G.1; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Pair two congruent triangles to form a rectangle. A triangle has half the area of that base-height rectangle.

Example 1: A triangular display has base 8 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.

$A = \frac{1}{2} \times 8 \times 7 = 28 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Example 2: A triangular display has base 6 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

$A = \frac{1}{2} \times 6 \times 4 = 12 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

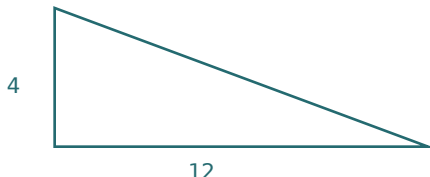
Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1–4

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

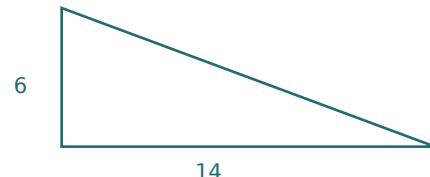
A triangular display has base 12 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 02

A triangular display has base 14 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 03

A triangular display has base 8 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 04

A triangular display has base 8 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.



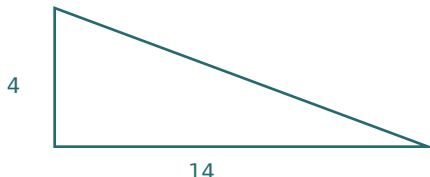
Explain your method.

Task cards 5–8

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

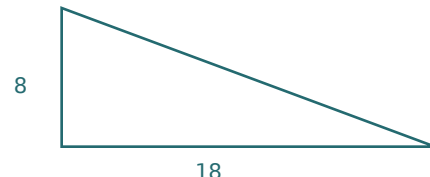
A triangular display has base 14 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 06

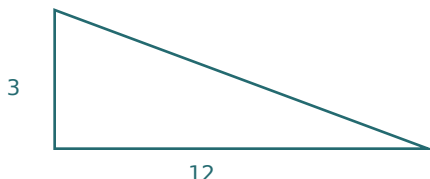
A triangular display has base 18 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 07

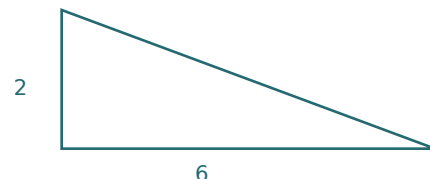
A triangular display has base 12 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 08

A triangular display has base 6 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.



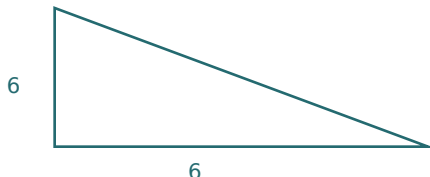
Explain your method.

Task cards 9–12

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

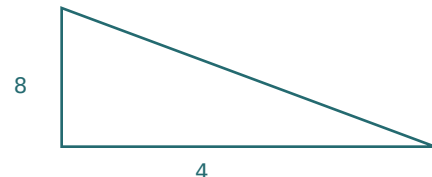
A triangular display has base 6 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 10

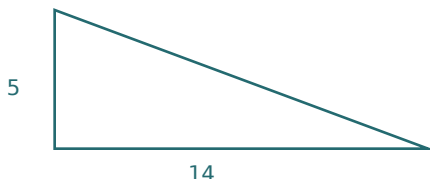
A triangular display has base 4 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 11

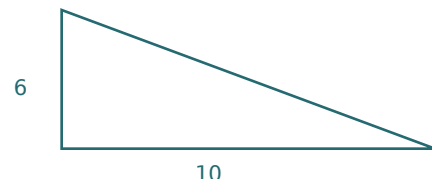
A triangular display has base 14 cm and perpendicular height 5 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 12

A triangular display has base 10 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.



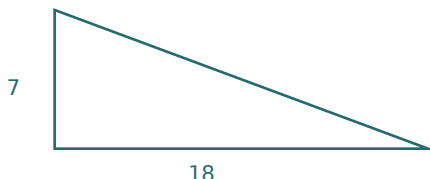
Explain your method.

Task cards 13–16

Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

A triangular display has base 18 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 14

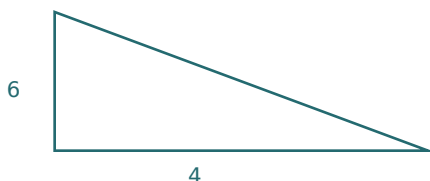
A triangular display has base 8 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 15

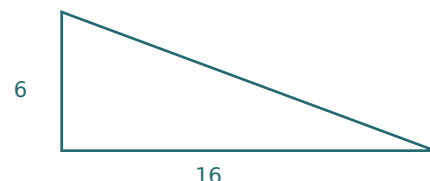
A triangular display has base 4 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

CARD 16

A triangular display has base 16 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.



Explain your method.

Task-card response record

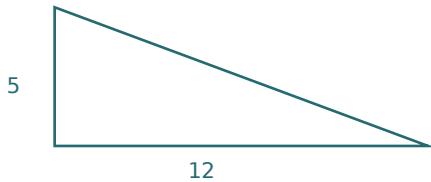
Use this page for final answers; attach drawings or full worked solutions on separate paper.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____

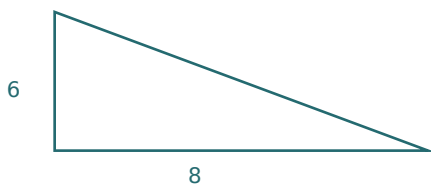
Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. A triangular display has base 12 cm and perpendicular height 5 cm. Find the area. Explain why multiplying base by height alone is too large.



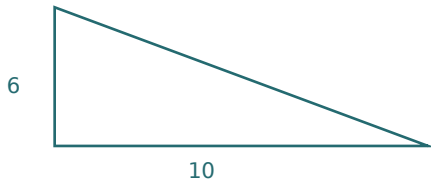
2. A triangular display has base 8 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.



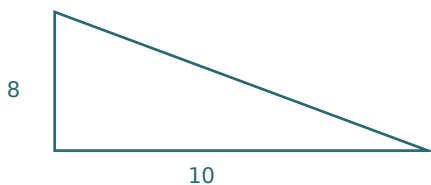
Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. A triangular display has base 10 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.



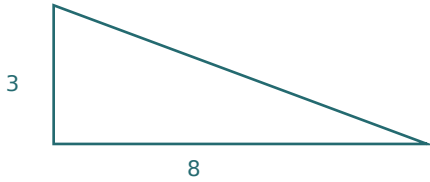
4. A triangular display has base 10 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.



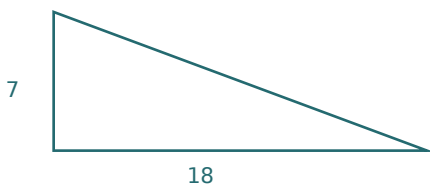
Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. A triangular display has base 8 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



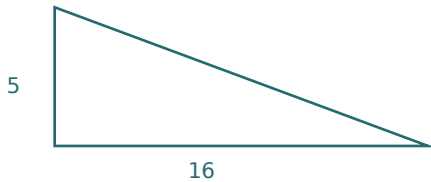
6. A triangular display has base 18 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.



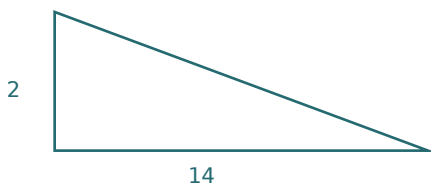
Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. A triangular display has base 16 cm and perpendicular height 5 cm. Find the area. Explain why multiplying base by height alone is too large.



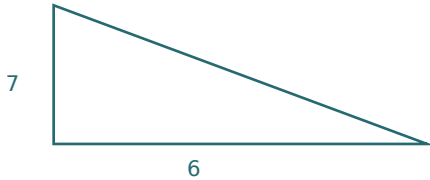
8. A triangular display has base 14 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.



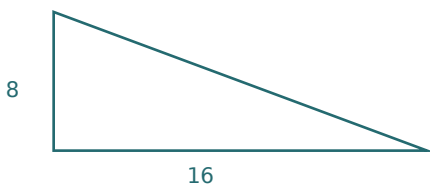
Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. A triangular display has base 6 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.



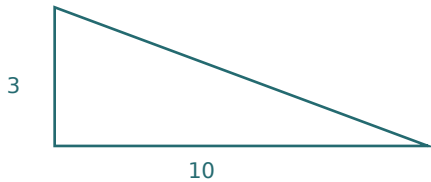
10. A triangular display has base 16 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.



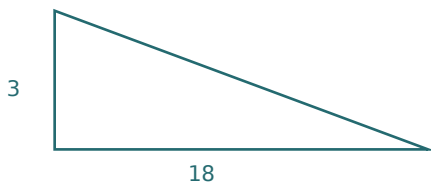
Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

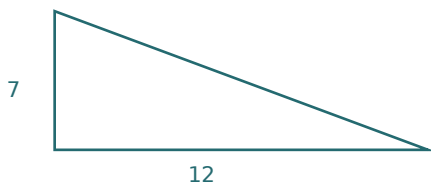
1. A triangular display has base 10 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



2. A triangular display has base 18 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



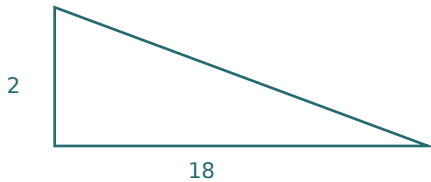
3. A triangular display has base 12 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.



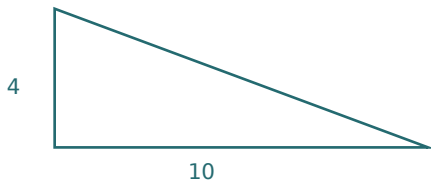
Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

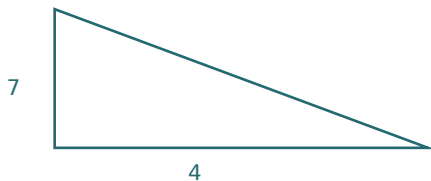
4. A triangular display has base 18 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.



5. A triangular display has base 10 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.



6. A triangular display has base 4 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. A triangular display has base 12 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

2. A triangular display has base 8 cm and perpendicular height 5 cm. Find the area. Explain why multiplying base by height alone is too large.

3. A triangular display has base 12 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

4. A triangular display has base 16 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.



Task-card worked solutions

1. 24 cm²

$A = \frac{1}{2} \times 12 \times 4 = 24 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 42 cm²

$A = \frac{1}{2} \times 14 \times 6 = 42 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 16 cm²

$A = \frac{1}{2} \times 8 \times 4 = 16 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 8 cm²

$A = \frac{1}{2} \times 8 \times 2 = 8 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

5. 28 cm²

$A = \frac{1}{2} \times 14 \times 4 = 28 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

6. 72 cm²

$A = \frac{1}{2} \times 18 \times 8 = 72 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Task-card worked solutions

7. 18 cm²

$A = \frac{1}{2} \times 12 \times 3 = 18 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

8. 6 cm²

$A = \frac{1}{2} \times 6 \times 2 = 6 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

9. 18 cm²

$A = \frac{1}{2} \times 6 \times 6 = 18 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

10. 16 cm²

$A = \frac{1}{2} \times 4 \times 8 = 16 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

11. 35 cm²

$A = \frac{1}{2} \times 14 \times 5 = 35 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

12. 30 cm²

$A = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Task-card worked solutions

13. 63 cm²

$A = \frac{1}{2} \times 18 \times 7 = 63 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

14. 8 cm²

$A = \frac{1}{2} \times 8 \times 2 = 8 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

15. 12 cm²

$A = \frac{1}{2} \times 4 \times 6 = 12 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

16. 48 cm²

$A = \frac{1}{2} \times 16 \times 6 = 48 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Warm-up worked solutions

1. 30 cm²

$A = \frac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 24 cm²

$A = \frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 30 cm²

$A = \frac{1}{2} \times 10 \times 6 = 30 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 40 cm²

$A = \frac{1}{2} \times 10 \times 8 = 40 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

5. 12 cm²

$A = \frac{1}{2} \times 8 \times 3 = 12 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

6. 63 cm²

$A = \frac{1}{2} \times 18 \times 7 = 63 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Warm-up worked solutions

7. 40 cm²

$A = \frac{1}{2} \times 16 \times 5 = 40 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

8. 14 cm²

$A = \frac{1}{2} \times 14 \times 2 = 14 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

9. 21 cm²

$A = \frac{1}{2} \times 6 \times 7 = 21 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

10. 64 cm²

$A = \frac{1}{2} \times 16 \times 8 = 64 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Classwork worked solutions

1. 15 cm²

$A = \frac{1}{2} \times 10 \times 3 = 15 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 27 cm²

$A = \frac{1}{2} \times 18 \times 3 = 27 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 42 cm²

$A = \frac{1}{2} \times 12 \times 7 = 42 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 18 cm²

$A = \frac{1}{2} \times 18 \times 2 = 18 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

5. 20 cm²

$A = \frac{1}{2} \times 10 \times 4 = 20 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

6. 14 cm²

$A = \frac{1}{2} \times 4 \times 7 = 14 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Exit-ticket worked solutions

1. 12 cm²

$A = \frac{1}{2} \times 12 \times 2 = 12 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 20 cm²

$A = \frac{1}{2} \times 8 \times 5 = 20 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 12 cm²

$A = \frac{1}{2} \times 12 \times 2 = 12 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 16 cm²

$A = \frac{1}{2} \times 16 \times 2 = 16 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A triangular display has base 10 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

2. A triangular display has base 6 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

3. A triangular display has base 12 cm and perpendicular height 5 cm. Find the area. Explain why multiplying base by height alone is too large.

4. A triangular display has base 18 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A triangular display has base 6 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.

6. A triangular display has base 18 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.

7. A triangular display has base 12 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.

8. A triangular display has base 10 cm and perpendicular height 5 cm. Find the area. Explain why multiplying base by height alone is too large.



Fall • Worked solutions

1. 20 cm²

$A = \frac{1}{2} \times 10 \times 4 = 20 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 6 cm²

$A = \frac{1}{2} \times 6 \times 2 = 6 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 30 cm²

$A = \frac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 63 cm²

$A = \frac{1}{2} \times 18 \times 7 = 63 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Fall • Worked solutions

5. 9 cm²

$A = \frac{1}{2} \times 6 \times 3 = 9 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

6. 54 cm²

$A = \frac{1}{2} \times 18 \times 6 = 54 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

7. 48 cm²

$A = \frac{1}{2} \times 12 \times 8 = 48 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

8. 25 cm²

$A = \frac{1}{2} \times 10 \times 5 = 25 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A triangular display has base 12 cm and perpendicular height 5 cm. Find the area. Explain why multiplying base by height alone is too large.

2. A triangular display has base 14 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.

3. A triangular display has base 16 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.

4. A triangular display has base 18 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A triangular display has base 8 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.

6. A triangular display has base 12 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

7. A triangular display has base 4 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

8. A triangular display has base 4 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



Winter • Worked solutions

1. 30 cm²

$A = \frac{1}{2} \times 12 \times 5 = 30 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 56 cm²

$A = \frac{1}{2} \times 14 \times 8 = 56 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 24 cm²

$A = \frac{1}{2} \times 16 \times 3 = 24 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 27 cm²

$A = \frac{1}{2} \times 18 \times 3 = 27 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Winter • Worked solutions

5. 32 cm²

$A = \frac{1}{2} \times 8 \times 8 = 32 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

6. 24 cm²

$A = \frac{1}{2} \times 12 \times 4 = 24 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

7. 8 cm²

$A = \frac{1}{2} \times 4 \times 4 = 8 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

8. 6 cm²

$A = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A triangular display has base 8 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.

2. A triangular display has base 4 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

3. A triangular display has base 18 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

4. A triangular display has base 4 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A triangular display has base 14 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

6. A triangular display has base 6 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.

7. A triangular display has base 16 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.

8. A triangular display has base 14 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.



Spring break • Worked solutions

1. 24 cm²

$A = \frac{1}{2} \times 8 \times 6 = 24 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 4 cm²

$A = \frac{1}{2} \times 4 \times 2 = 4 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 18 cm²

$A = \frac{1}{2} \times 18 \times 2 = 18 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 6 cm²

$A = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.



Spring break • Worked solutions

5. 28 cm^2

$A = \frac{1}{2} \times 14 \times 4 = 28 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

6. 24 cm^2

$A = \frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

7. 64 cm^2

$A = \frac{1}{2} \times 16 \times 8 = 64 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

8. 56 cm^2

$A = \frac{1}{2} \times 14 \times 8 = 56 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A triangular display has base 4 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.

2. A triangular display has base 4 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.

3. A triangular display has base 12 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.

4. A triangular display has base 8 cm and perpendicular height 2 cm. Find the area. Explain why multiplying base by height alone is too large.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A triangular display has base 16 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.

6. A triangular display has base 10 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.

7. A triangular display has base 12 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.

8. A triangular display has base 4 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.



Summer • Worked solutions

1. 4 cm²

$A = \frac{1}{2} \times 4 \times 2 = 4 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 6 cm²

$A = \frac{1}{2} \times 4 \times 3 = 6 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 18 cm²

$A = \frac{1}{2} \times 12 \times 3 = 18 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 8 cm²

$A = \frac{1}{2} \times 8 \times 2 = 8 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Summer • Worked solutions

5. 56 cm^2

$A = \frac{1}{2} \times 16 \times 7 = 56 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

6. 40 cm^2

$A = \frac{1}{2} \times 10 \times 8 = 40 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

7. 18 cm^2

$A = \frac{1}{2} \times 12 \times 3 = 18 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

8. 8 cm^2

$A = \frac{1}{2} \times 4 \times 4 = 8 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use decomposition to find triangle area. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Pair two congruent triangles to form a rectangle. A triangle has half the area of that base-height rectangle.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A triangular display has base 8 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.

2. A triangular display has base 18 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.

3. A triangular display has base 6 cm and perpendicular height 6 cm. Find the area. Explain why multiplying base by height alone is too large.

4. A triangular display has base 12 cm and perpendicular height 7 cm. Find the area. Explain why multiplying base by height alone is too large.



Substitute lesson • Worked solutions

1. 28 cm²

$A = \frac{1}{2} \times 8 \times 7 = 28 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 54 cm²

$A = \frac{1}{2} \times 18 \times 6 = 54 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 18 cm²

$A = \frac{1}{2} \times 6 \times 6 = 18 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 42 cm²

$A = \frac{1}{2} \times 12 \times 7 = 42 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use decomposition to find triangle area. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Pair two congruent triangles to form a rectangle. A triangle has half the area of that base-height rectangle.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A triangular display has base 10 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.

2. A triangular display has base 8 cm and perpendicular height 8 cm. Find the area. Explain why multiplying base by height alone is too large.

3. A triangular display has base 8 cm and perpendicular height 4 cm. Find the area. Explain why multiplying base by height alone is too large.

4. A triangular display has base 10 cm and perpendicular height 3 cm. Find the area. Explain why multiplying base by height alone is too large.



Emergency substitute lesson • Worked solutions

1. 15 cm^2

$A = \frac{1}{2} \times 10 \times 3 = 15 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

2. 32 cm^2

$A = \frac{1}{2} \times 8 \times 8 = 32 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

3. 16 cm^2

$A = \frac{1}{2} \times 8 \times 4 = 16 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

4. 15 cm^2

$A = \frac{1}{2} \times 10 \times 3 = 15 \text{ cm}^2$. The base-height rectangle contains two congruent triangles.

A completed lesson-plan example

Level and focus

Grade 6; NC.6.G.1; Use decomposition to find triangle area.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.6.G.1 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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